EE1007
Introduction to Computer Science Laboratory

Lab 5
Control Statement III (repeat control)

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Lab 5.1
Print Diamond

- In this problem, you need to write a program which can print a hollow diamond shape on the screen.
- The program will show the prompts to guide the user to enter a number, take a non-zero integer from keyboard, and then print the hollow diamond by using character "*".
- The number of "*" used (i.e., the size of the hollow diamond) is decided by the user-provided integer (see next pages for examples)
- **The range of user-provided integer is [3, 2^{31}-1]**
- The form of diamond must follow the examples (see next pages)

Lab 5.1
Print Diamond

- The screen dialog should appear as follows:

```
Enter the size of the diamond: 2
Enter again: 5
***  ***
**    **
*      *
**    **
***  ***

Process returned 0 (0x0)   execution time : 7.753 s
Press any key to continue.
```


Lab 5.1
Print Diamond

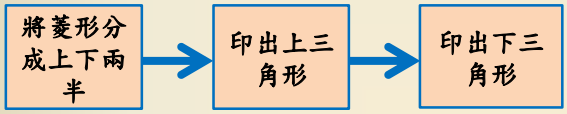
- Hints: The number of space in each level is just like the graph on the left, but in a reverse version (hollow diamond). In this example, size=9.

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

```
***** *****
****   ****
***     ***
**       **
*         *
**       **
***     ***
****   ****
***** *****
```

Lab 5.1
Print Diamond

◆思考方向：



Lab 5.1
Print Diamond

◆Step 1. 將菱形分成上下兩半：

```
cout << "Enter the size of the daimond: ";
cin >> size;
rows = size / 2 + 1;
```

Lab 5.1
Print Diamond

◆Step 2. 印出上三角形：

```
//upper half
for(int i = 1; i <= rows; i++){
    for(int j = 0; j <= rows - i; j++)
    {
        cout << "*";
    }
    for(int j = 1; j <= i * 2 - 1; j++){
        cout << " " ;
    }
    for(int j = 0; j <= rows - i; j++){
        cout << "*" ;
    }
    cout << endl;
}
```



For迴圈的條件：思考總共列數和目前在第幾列的關係

Lab 5.1
Print Diamond

◆Step 3. 印出下三角形：

```
//bottom half
for(int i = 1; i <= rows - 1; i++){
    for(int j = 0; j <= i; j++)
    {
        cout << "*";
    }
    for(int j = 1; j <= (rows - i) * 2 - 1; j++){
        cout << " " ;
    }
    for(int j = 0; j <= i; j++)
    {
        cout << "*";
    }
    cout << endl;
}
```



For迴圈的條件：思考總共列數和目前在第幾列的關係

Lab5.2
Pyramid

- In this problem, you are asked to write a program which can show a numerical pyramid with basic arithmetic sequence.
- Your program will take a non-zero integer which is used to determine the height of the pyramid from keyboard, and then print the pyramid.
- The pyramid is composed of a basic arithmetic sequence in increasing order.
- The output should print each number with 3-bits width clearly with one space between each number.
- **The range of user-provided integer is [0, 2^{31}-1]**

Lab5.2
Pyramid

- The screen dialog should appear as follows:

```
Enter number of rows: 11
                                  1
                              2   3   4
                          5   6   7   8   9
                     10  11  12  13  14  15  16
                 17  18  19  20  21  22  23  24  25
             26  27  28  29  30  31  32  33  34  35  36
         37  38  39  40  41  42  43  44  45  46  47  48  49
     50  51  52  53  54  55  56  57  58  59  60  61  62  63  64
 65  66  67  68  69  70  71  72  73  74  75  76  77  78  79  80  81
82  83  84  85  86  87  88  89  90  91  92  93  94  95  96  97  98  99 100
101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121

Process returned 0 (0x0)   execution time : 1.395 s
Press any key to continue.
```

Lab5.2
Pyramid

◆思考方式：



Lab5.2
Pyramid

◆Step 1. 計算目前在第幾層：

```
for(int i = 1; i <= rows; i++)
{
    //
    //
    //
}
```

Lab5.2
Pyramid

◆Step 2. 給出對應的空格位移：

```
for(int j = 1; j <= rows-i; j++)
{
    cout << "    ";
}
```

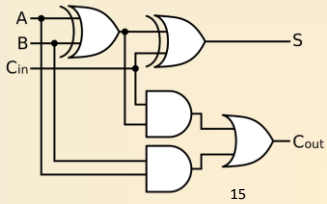

Lab5.2
Pyramid

◆Step 3. 計算當前要印出的數值大小：

```
for(int j = 1; j <= i * 2 - 1; j++){
    cnt++;
    cout << setw(4) << cnt;
}
cout << endl;
```

Lab 5.3
Full Adder

- You should just learn the logic gates in DL course. The circuit bellowed is a full adder.
- In this problem, you are asked to write a program for simulating the function of a full adder.
  - https://youtu.be/GuDPXbGiRHk
  - Start from 9:28



Lab 5.3
Full Adder

- The relation between the inputs and the outputs of a full adder is like the table shown bellowed.
- In your program, you need to print specific row of the truth table based on the corresponding input values.

Table 4.4
Full Adder

x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Lab 5.3
Full Adder

- A full adder takes three inputs (A, B, Cin) and generate two outputs(Sum, Cout).
- Each inputs and outputs is restricted to 0 and 1. If the input value is not 0 or 1, your program will show error message and ask the user to enter the inputs again.
- The program will continuous execution until it obtain -1 from input A.
- You **MUST** use **logic operators** in your program, others you will get 0 point.
  - i.e., you must use && || !

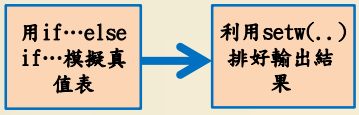
Lab 5.3
Full Adder

- The screen dialog should appear as follows:

```
Enter a: 1
Enter b: 0
Enter Cin: 1
-------------------
  A  B  Cin Sum Cout
  1  0    1   0    1
-------------------
Enter a: -5
Enter b: 1
Enter Cin: 1
!!! Enter the inputs again !!!
Enter a: 1
Enter b: 1
Enter Cin: 0
-------------------
  A  B  Cin Sum Cout
  1  1    0   0    1
-------------------
Enter a: -1
Bye~
Process returned 0 (0x0)   execution time : 10.852 s
Press any key to continue.
```

Lab 5.3
Full Adder

◆思考方式：



Lab 5.3
Full Adder

◆Step 1. 用if…else if…模擬真值表：

```
//logic
if (a == 0 && b== 0 && ci == 0){//000
    sum = 0;
    co = 0;
}
else if (a == 0 && b== 0 && ci == 1){//001
    sum = 1;
    co = 0;
}
else if (a == 0 && b== 1 && ci == 0){//010
    sum = 1;
    co = 0;
}
else if (a == 0 && b== 1 && ci == 1){//011
    sum = 0;
    co = 1;
}
else if (a == 1 && b== 0 && ci == 0){//100
    sum = 1;
    co = 0;
}
```

```
else if (a == 1 && b== 0 && ci == 1){//101
    sum = 0;
    co = 1;
}
else if (a == 1 && b== 1 && ci == 0){//110
    sum = 0;
    co = 1;
}
else if (a == 1 && b== 1 && ci == 1){//111
    sum = 1;
    co = 1;
}
else {
    cout << "!!! Enter the inputs again !!!" << endl;
}
```

Lab 5.3
Full Adder

◆Step 2. 利用setw(..)排好輸出結果：

```
cout << "--------------------" << endl;
cout << setw(3) << "A" << setw(3) << "B" << setw(5) << "Cin" << setw(5) << "Sum " << setw(4) << "Cout" << endl;
cout << setw(3) << a << setw(3) << b << setw(5) << ci << setw(4) << sum << setw(5) << co << endl;
cout << "--------------------" << endl;
```


Evaluation

◆Lab5.1(30%)
- ➢ 5 testcases, 6% each

◆Lab5.2(30%)
- ➢ 5 testcases, 6% each

◆Lab5.3(40%)
- ➢ 5 testcases, 8% each